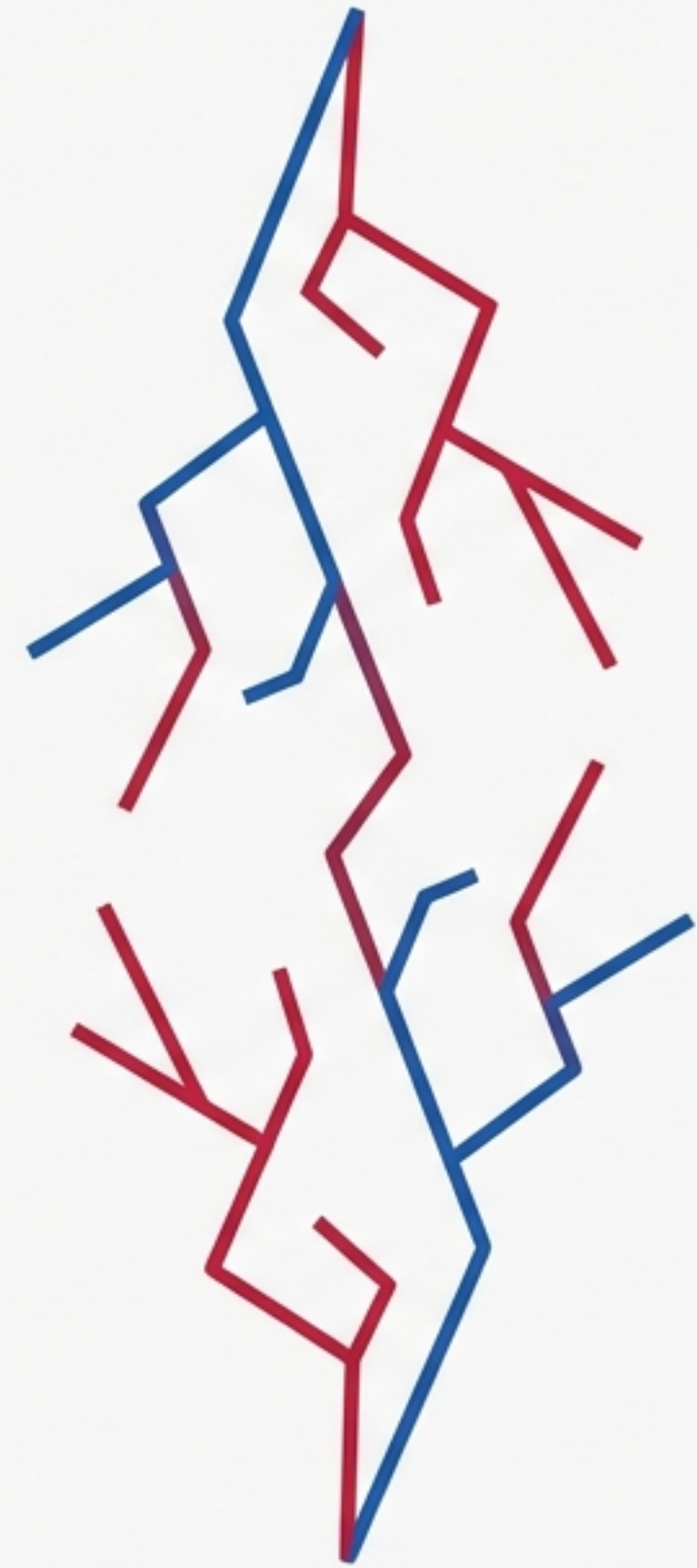


Electrostatics: From Fundamental Particles to Field Theory

A structural analysis of Electric Charges, Forces, and Fields

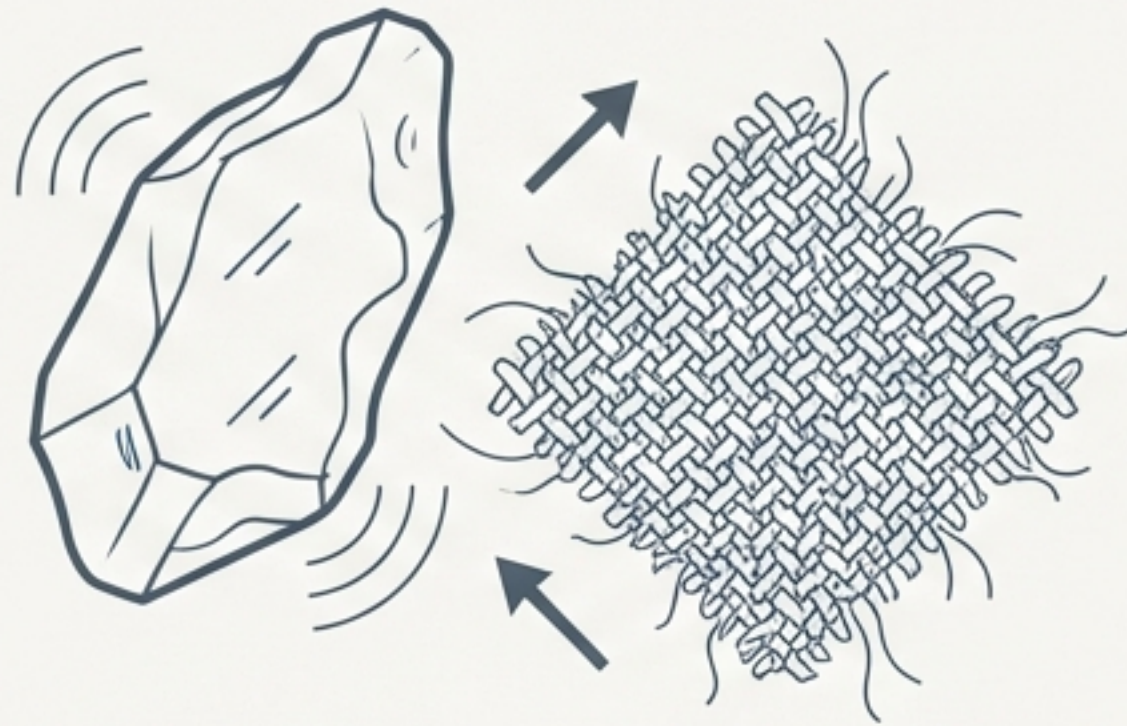
Static means anything that does not move or change with time. **Electrostatics** deals with the study of forces, fields, and potentials arising from static charges.



Context: The audible crackle of synthetic clothes or the shock from a car door are everyday manifestations of accumulated electric charges.

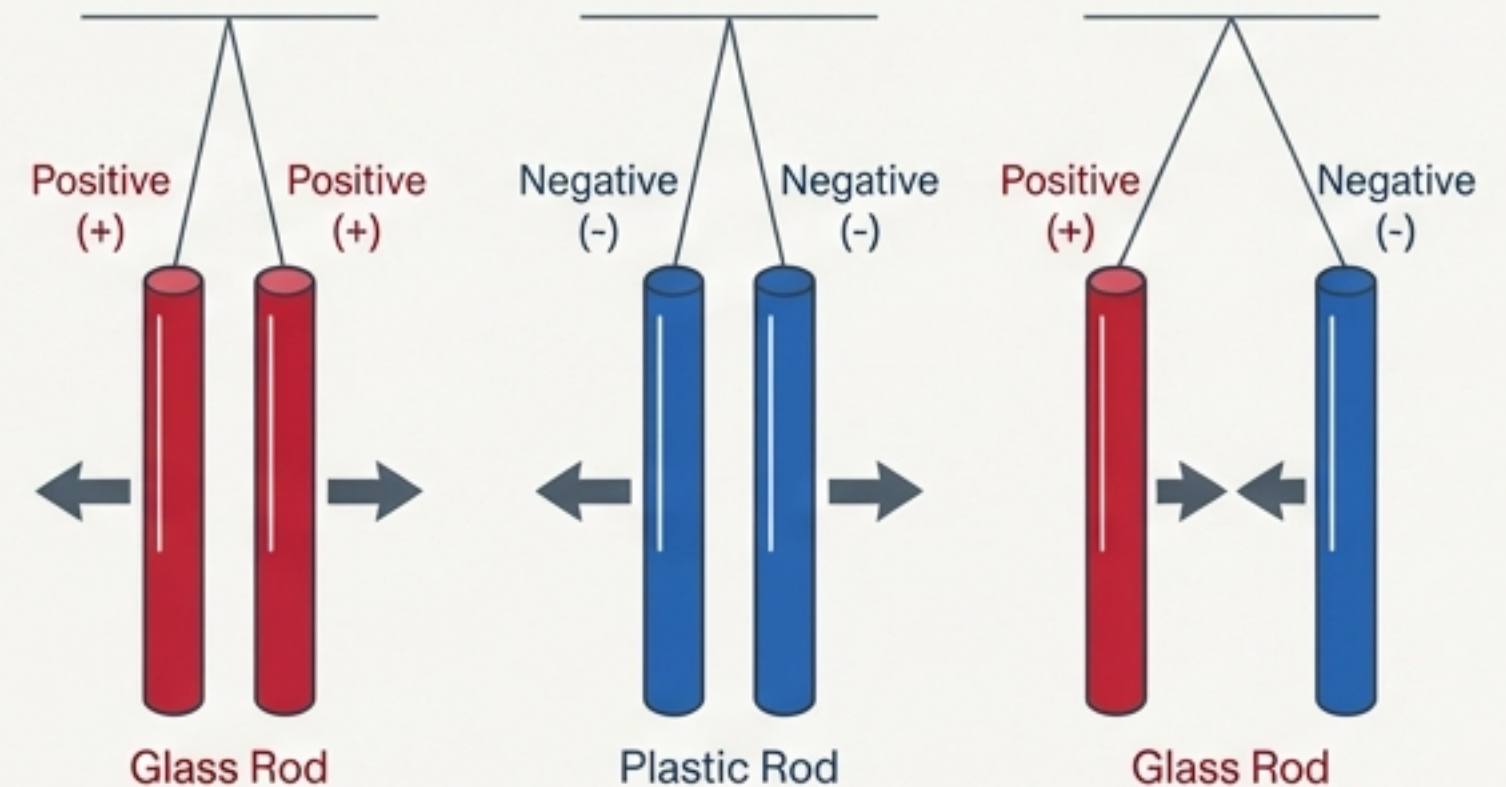
The Discovery of Charge

Historical Origins



Thales of Miletus (600 BC) observed that amber rubbed with wool attracts light objects. “Electricity” is derived from “elektron”, the Greek word for amber.

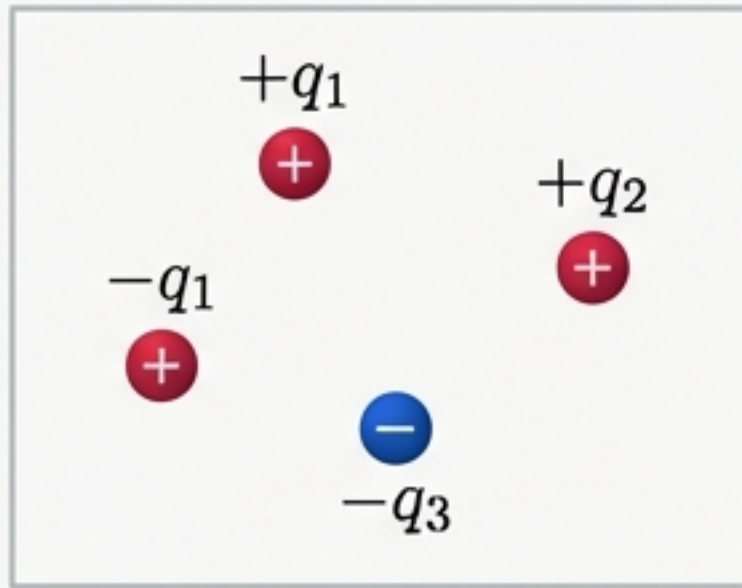
The Binary Nature of Charge



- Conductors: Allow free charge movement (e.g., Metals, Earth).
- Insulators: High resistance; charges stay fixed (e.g., Glass, Nylon).

Fundamental Properties of Electric Charge

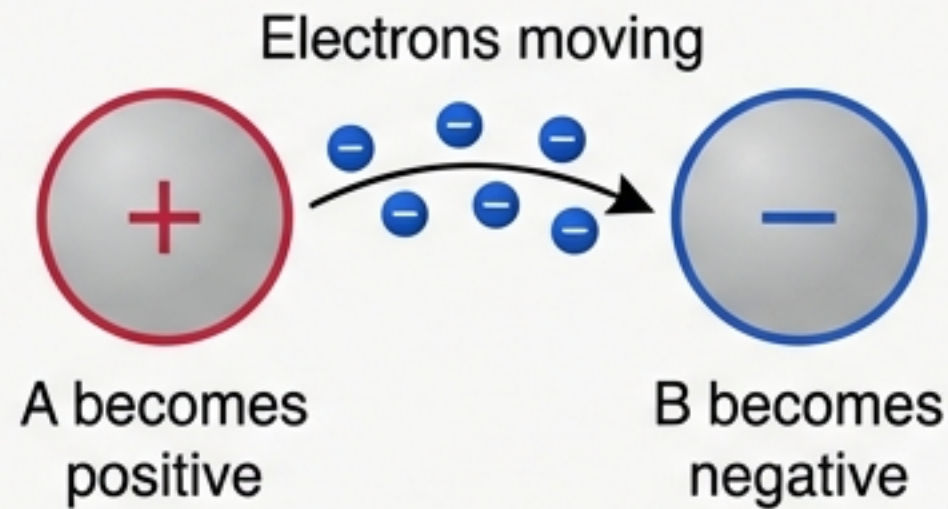
Additivity



$$Q_{total} = q_1 + q_2 - q_3$$

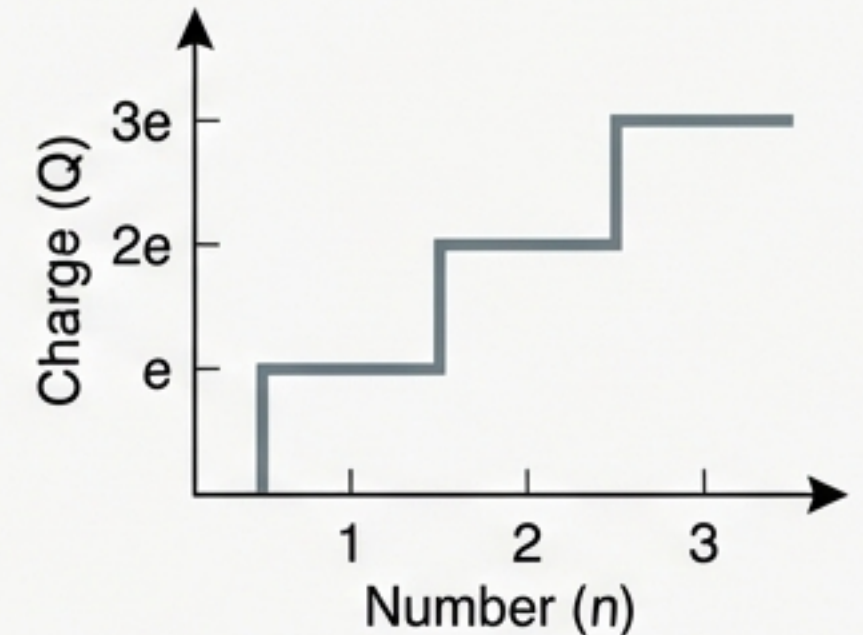
Charges are scalars. The total charge is the algebraic sum of individual point charges.

Conservation



In an isolated system, total charge is conserved. Charges are transferred, never created or destroyed.

Quantization



Charge exists in discrete packets. $q = ne$ where n is an integer.

Elementary Unit: $e = 1.602 \times 10^{-19}$ C.

Significance: At macroscopic scales (μC), charge appears continuous. At microscopic scales, the "grainy" nature is dominant (Millikan's discovery).

Coulomb's Law: Quantifying the Interaction

Electrostatic Force

$$F = k \frac{|q_1 q_2|}{r^2}$$

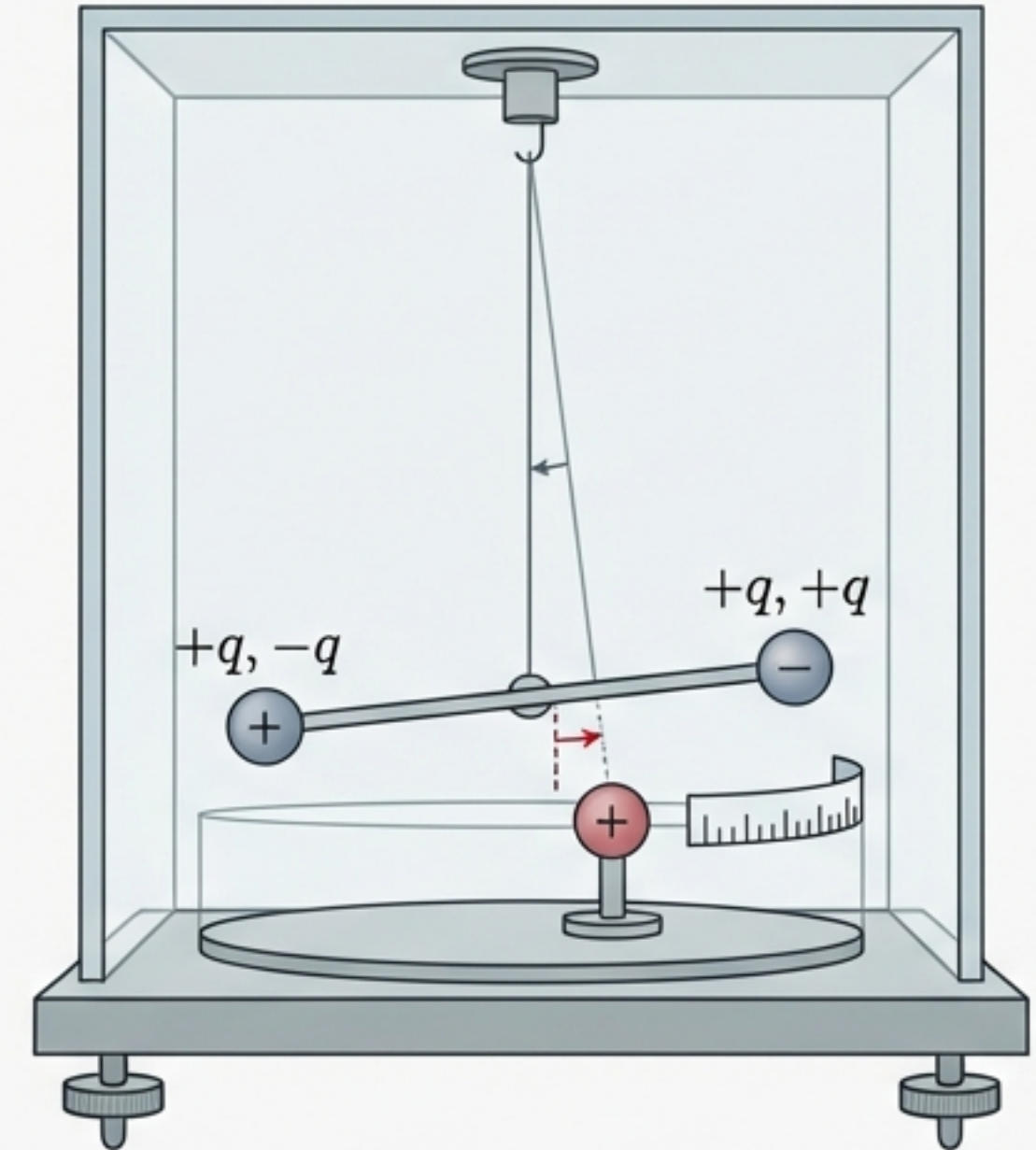
Proportionality Constant
($9 \times 10^9 \text{ Nm}^2\text{C}^{-2}$)

Product of Charges

Inverse Square Law

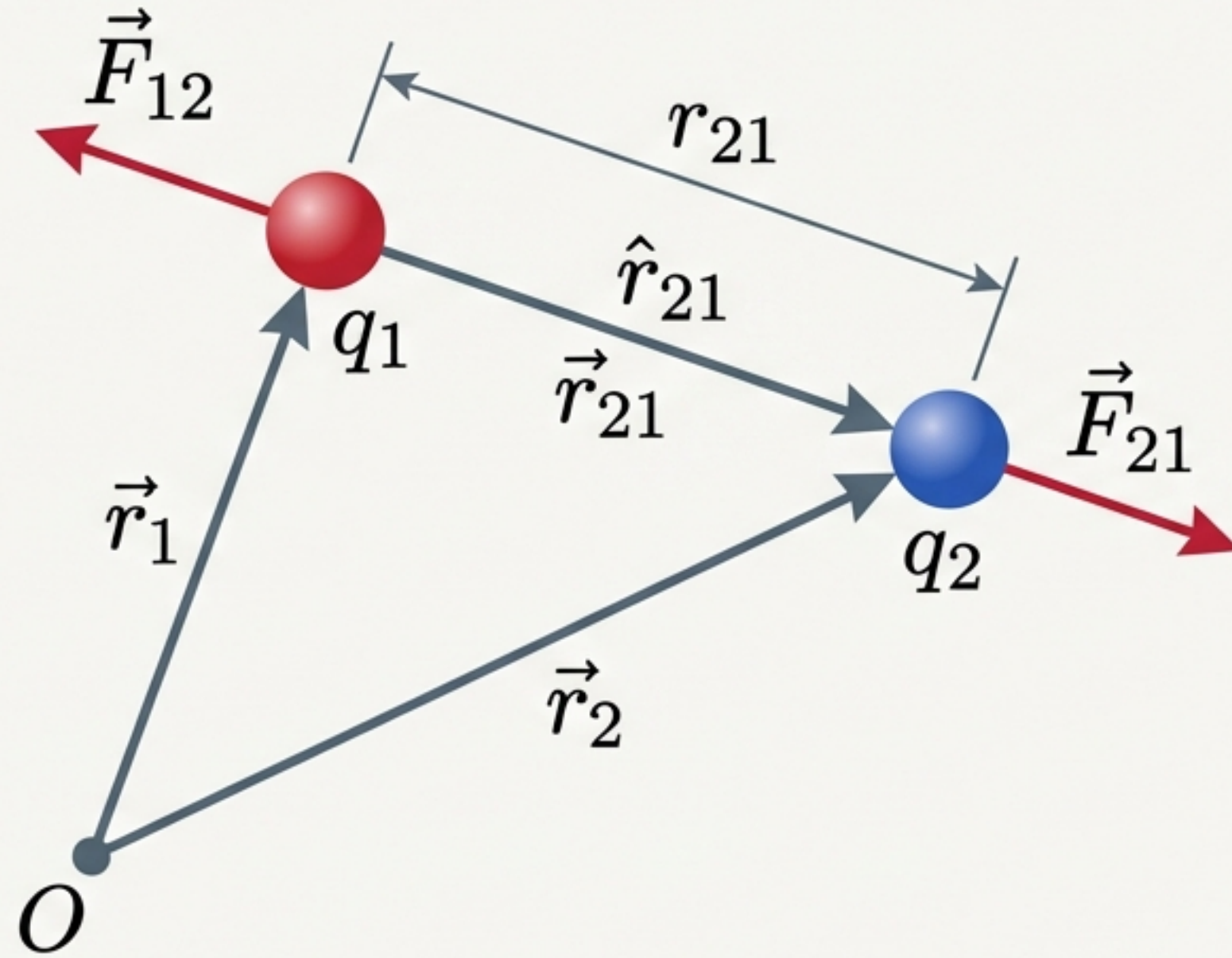
$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$$

(Permittivity of free space)



Coulomb's Torsion Balance: Used to measure feeble electrostatic forces.

Vector Notation and Directionality

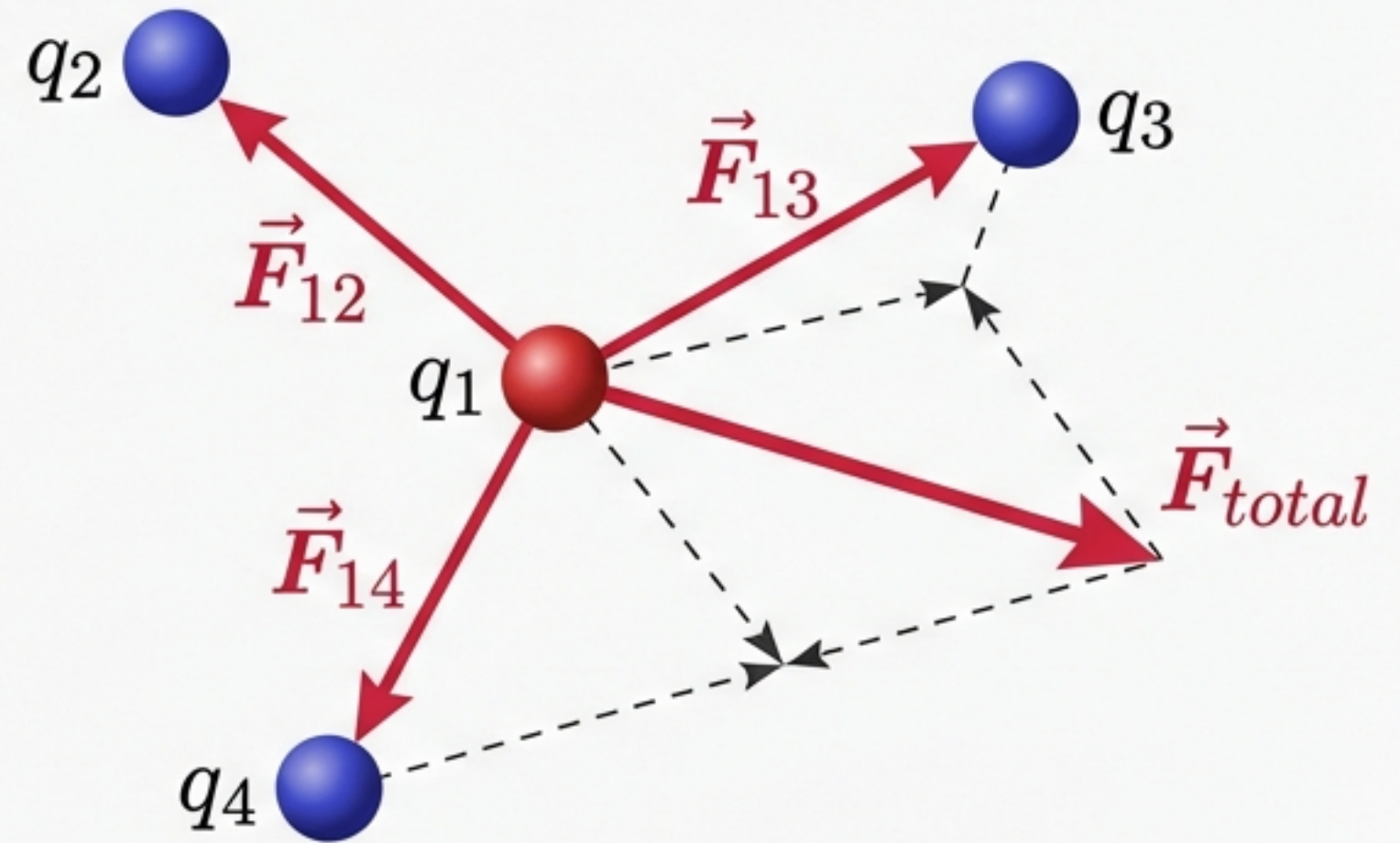


$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{21}^2} \hat{r}_{21}$$

1. $\hat{r}_{21}$: Unit vector pointing from 1 to 2.
2. Repulsion: If $q_1 q_2 > 0$ (Like charges).
3. Attraction: If $q_1 q_2 < 0$ (Unlike charges).
4. Newton's 3rd Law: $\vec{F}_{12} = -\vec{F}_{21}$ (Action-Reaction Pair).

The Principle of Superposition

Coulomb's law explains the interaction between two charges. In a system of multiple charges, the force on any single charge is the vector sum of all forces exerted by the others.



$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \dots + \vec{F}_{1n} = \frac{q_1}{4\pi\epsilon_0} \sum_{i=2}^n \frac{q_i}{r_{1i}^2} \hat{r}_{1i}$$

The Paradigm Shift: Introducing the Electric Field

The Question

Action-at-a-Distance: How does charge q_2 “know” q_1 is there? Does the force act instantaneously across empty space?

$$\vec{E} = \lim_{q \rightarrow 0} \left(\frac{\vec{F}}{q} \right)$$

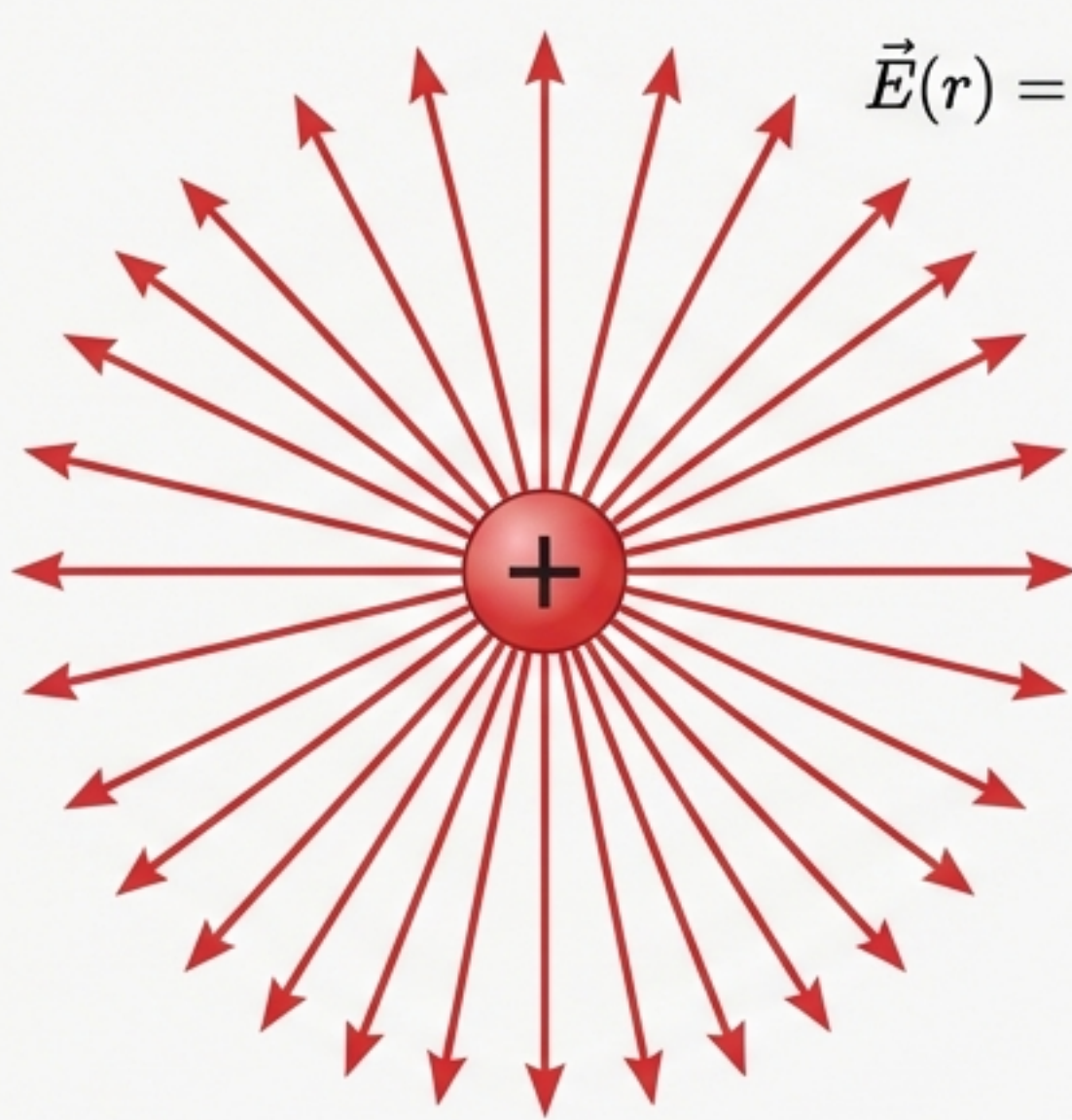
Force per unit
Charge (N/C)

The Answer: Field Theory

A source charge Q modifies the space around it, creating an Electric Field (E). A test charge q placed in this modified space interacts with the field, not directly with the distant source.

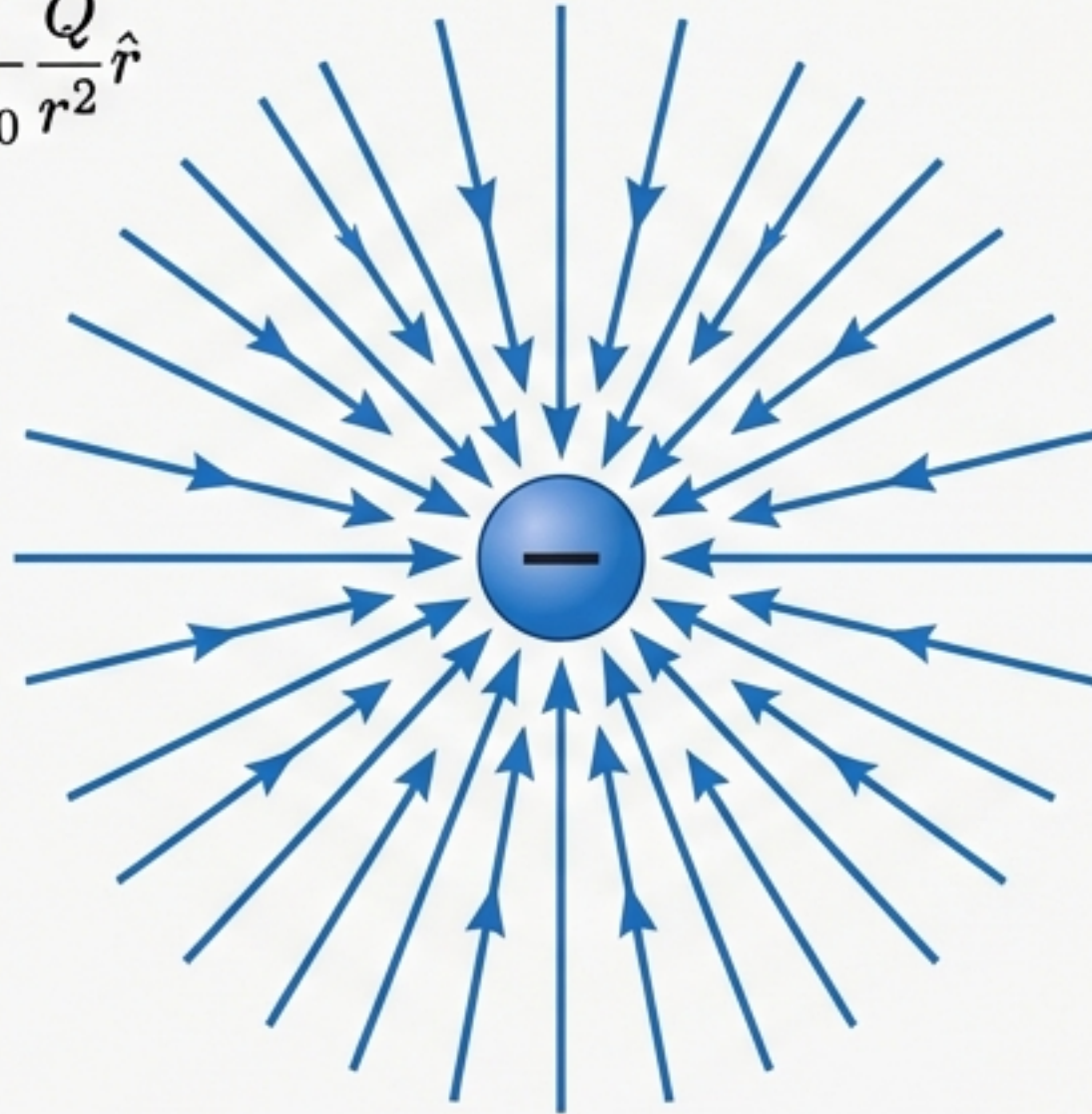
Physical Significance: The field carries energy and propagates at the speed of light (c), solving the time-delay problem.

Electric Field of a Point Charge



Positive Charge ($Q > 0$)

$$\vec{E}(r) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r}$$



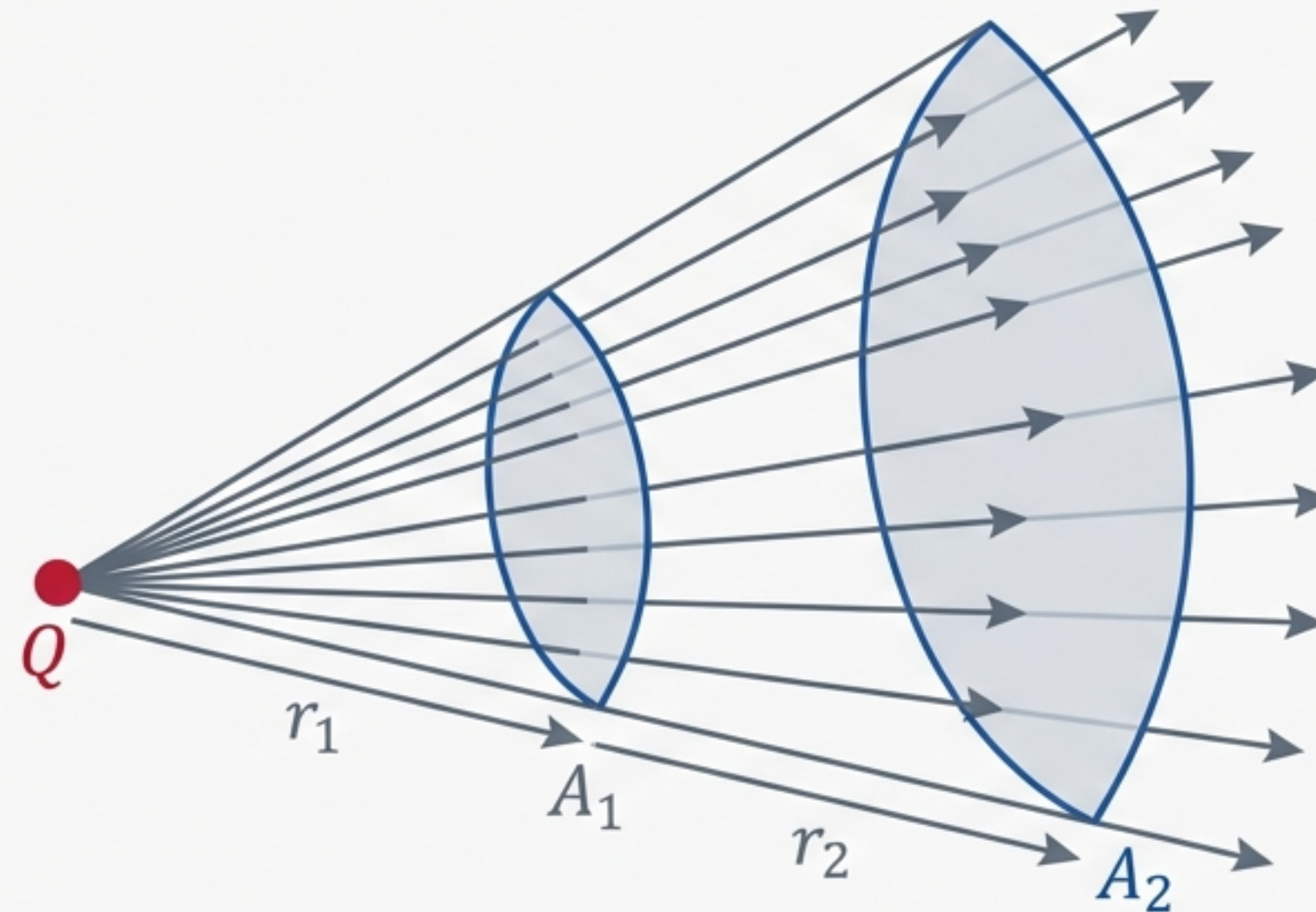
Negative Charge ($Q < 0$)

- Key Properties
 - Spherical Symmetry: Magnitude depends only on distance r .
 - Inverse Square Law: Field strength diminishes as $1/r^2$.
 - Superposition: Net field is the vector sum of individual fields.

Visualizing the Invisible: Electric Field Lines

Michael Faraday's intuitive method for mapping fields.

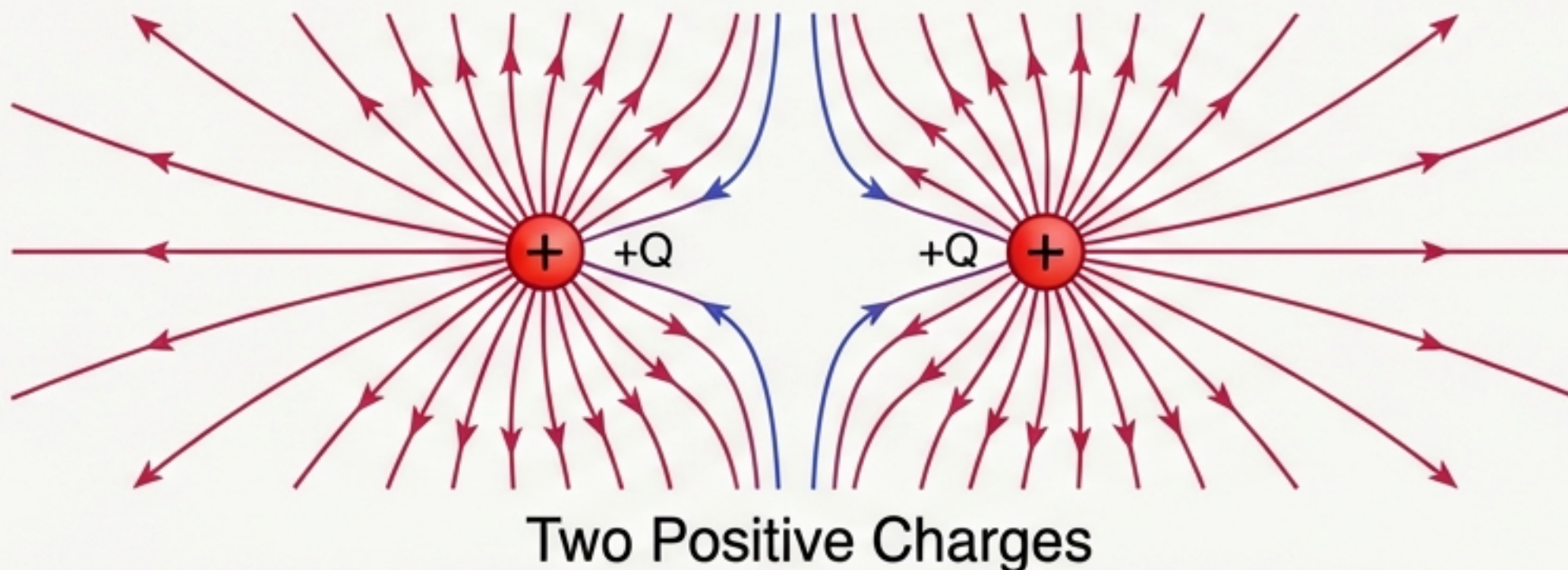
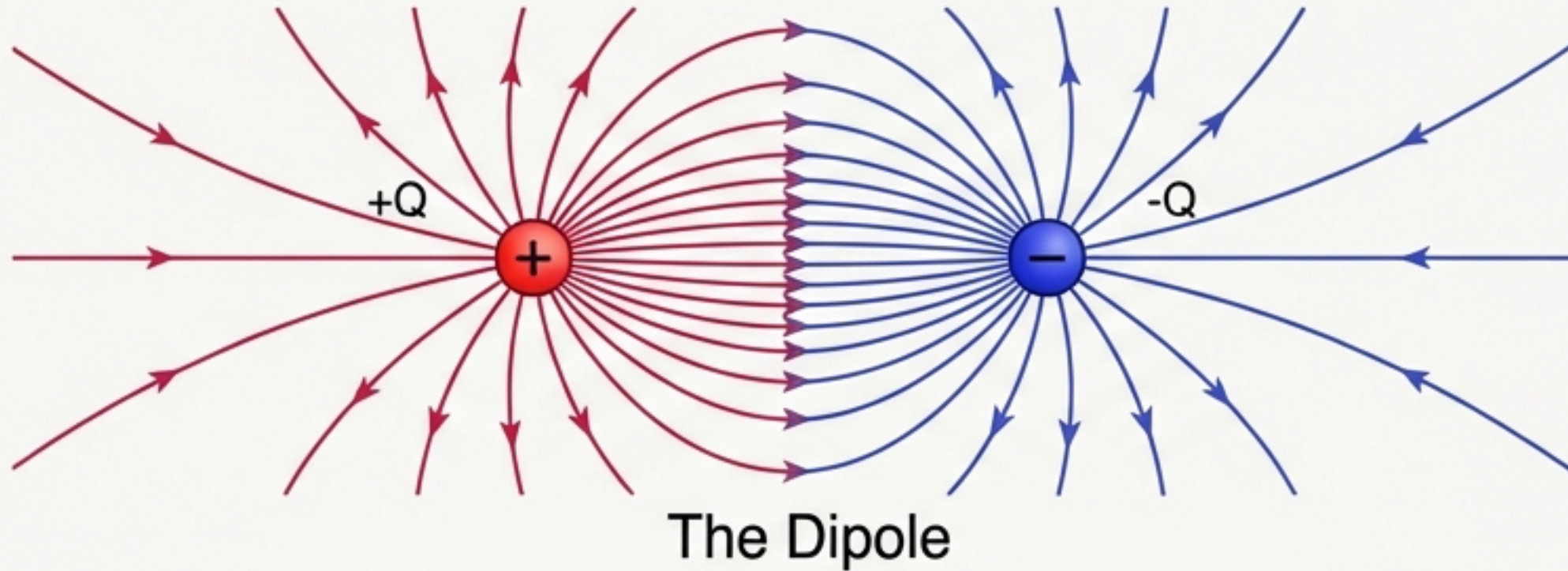
Tangent = Direction:
The tangent to the curve at any point gives the direction of the field.



Density = Magnitude:
Crowded lines indicate a strong field; widely spaced lines indicate a weak field.

Area Dependence: Number of lines is constant, but area scales as r^2 , confirming the $1/r^2$ field strength relationship.

Properties and Configurations of Field Lines

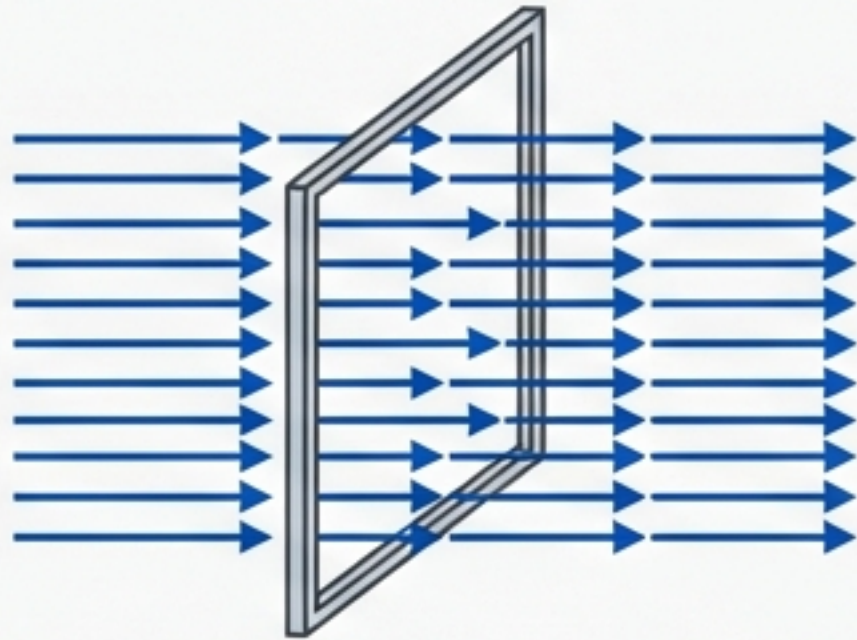


The Four Rules

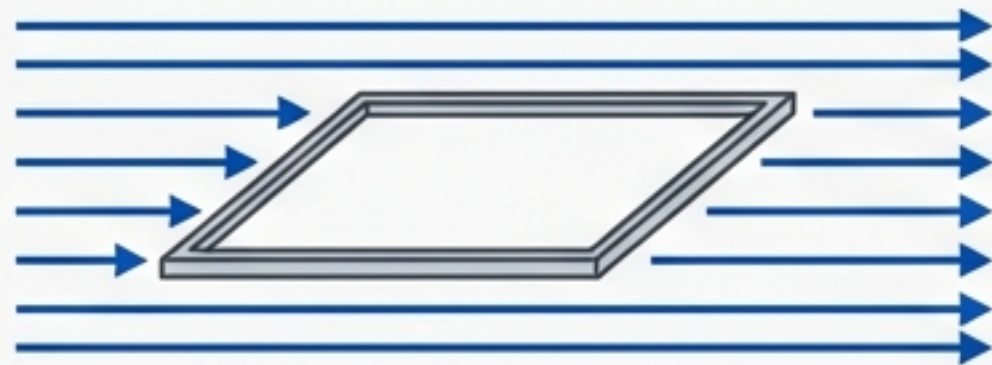
- **Start/End:** Positive $\rightarrow$ Negative (or Infinity).
- **Continuity:** No breaks in charge-free regions.
- **No Intersections:** Lines never cross (would imply two directions at one point).
- **No Closed Loops:** Electrostatic fields are conservative.

The Concept of Electric Flux

Water Analogy

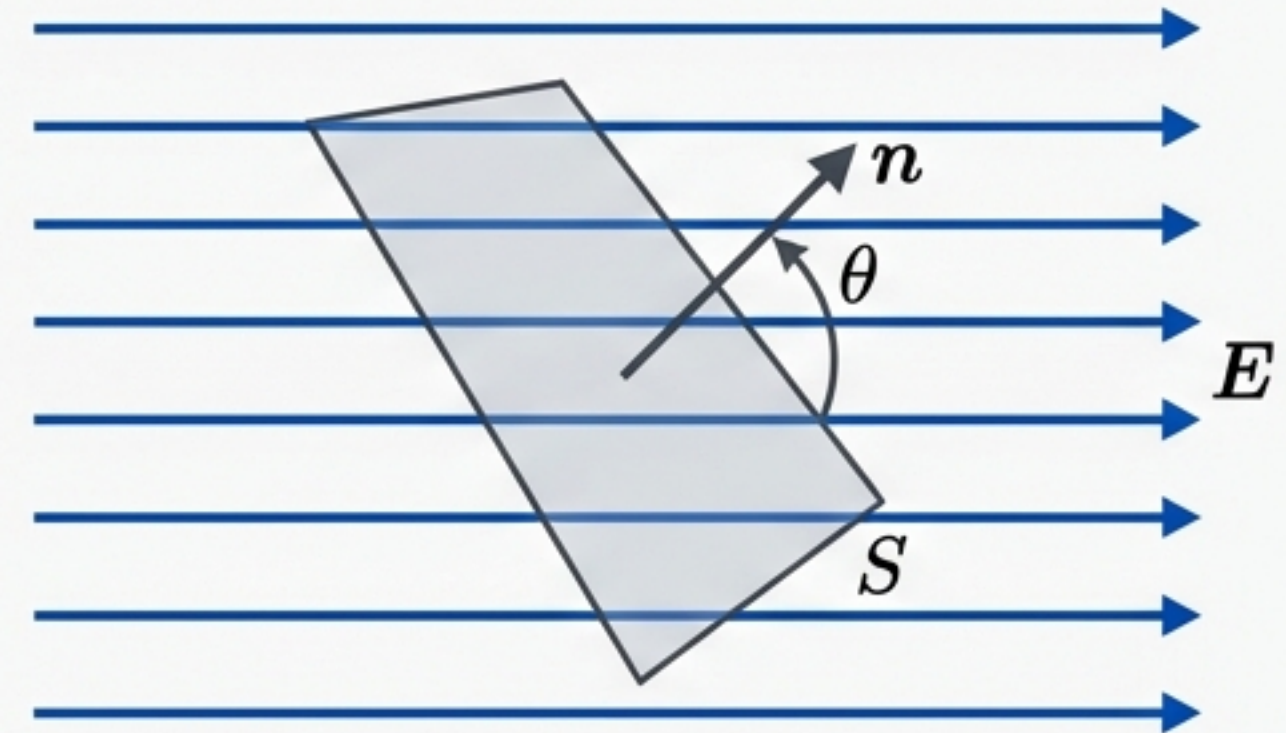


Frame Perpendicular = Max Flow



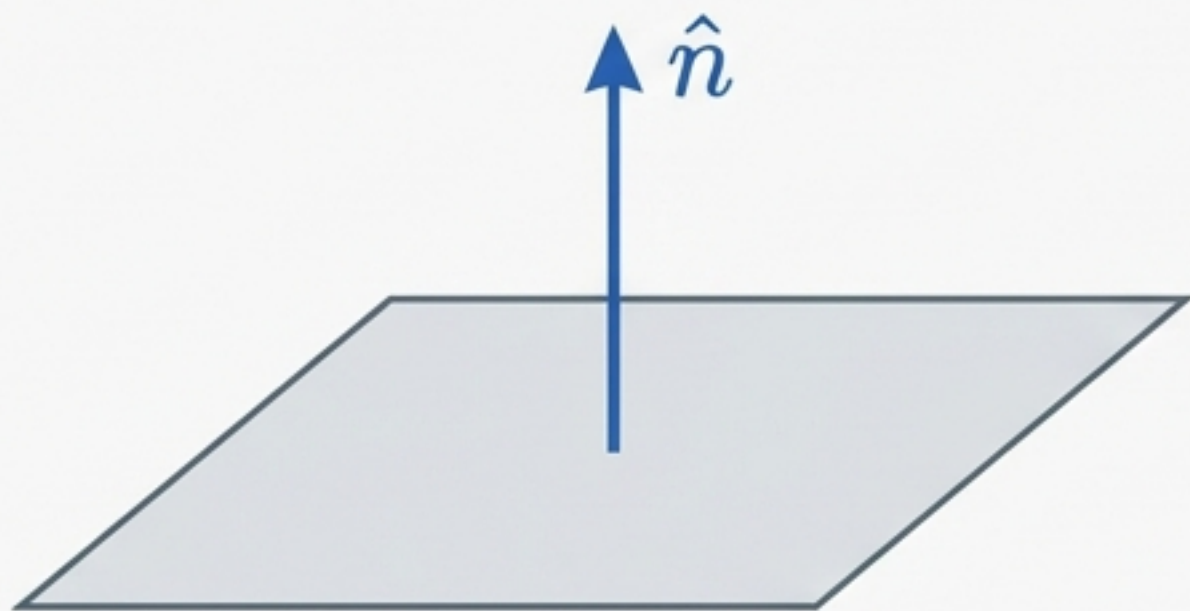
Frame Parallel = Zero Flow

Electrostatic Equivalent



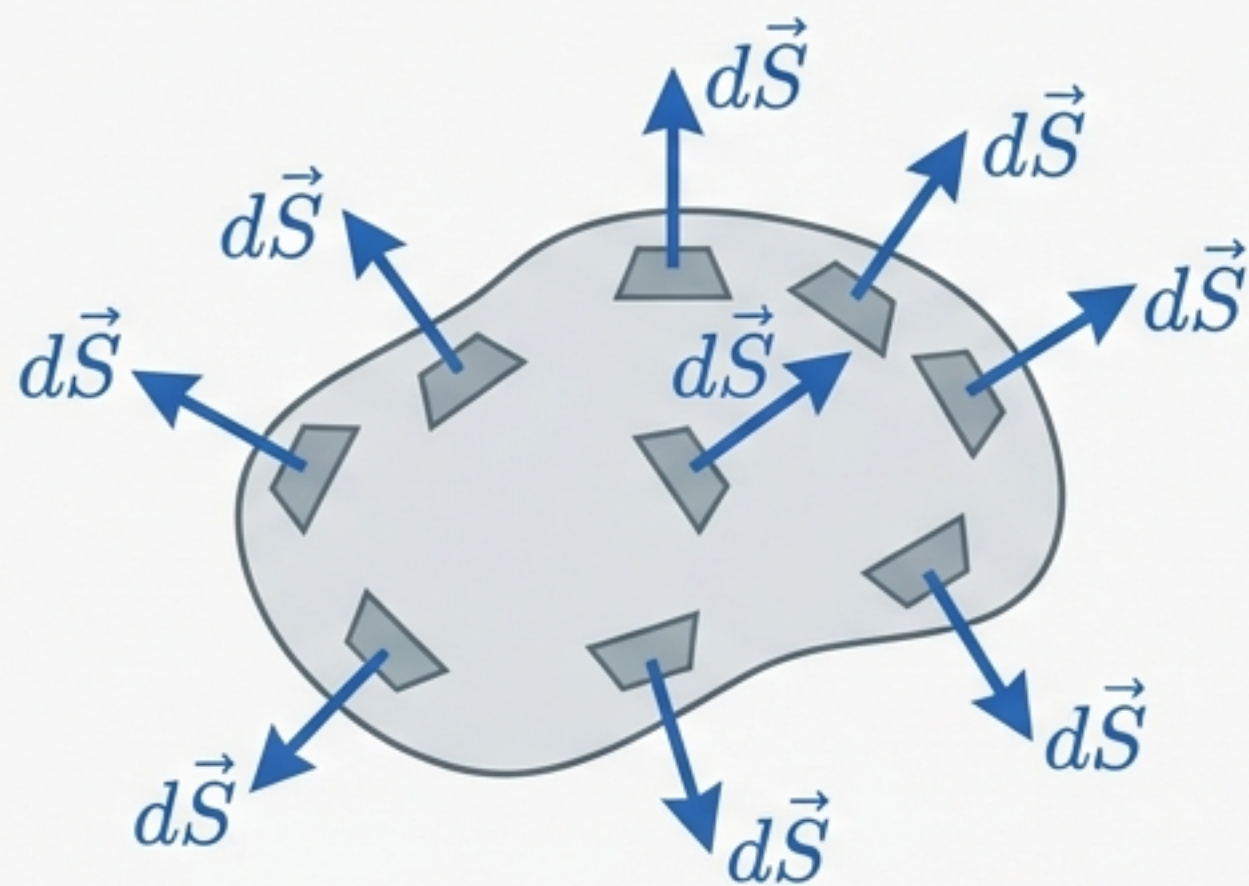
Flux (Φ) is proportional to the number of field lines crossing a surface.

- It depends on three factors: Field Strength (E), Surface Area (S), and Orientation (θ).



The Area Vector

To calculate flux mathematically, "Area" must be treated as a vector, not a scalar.



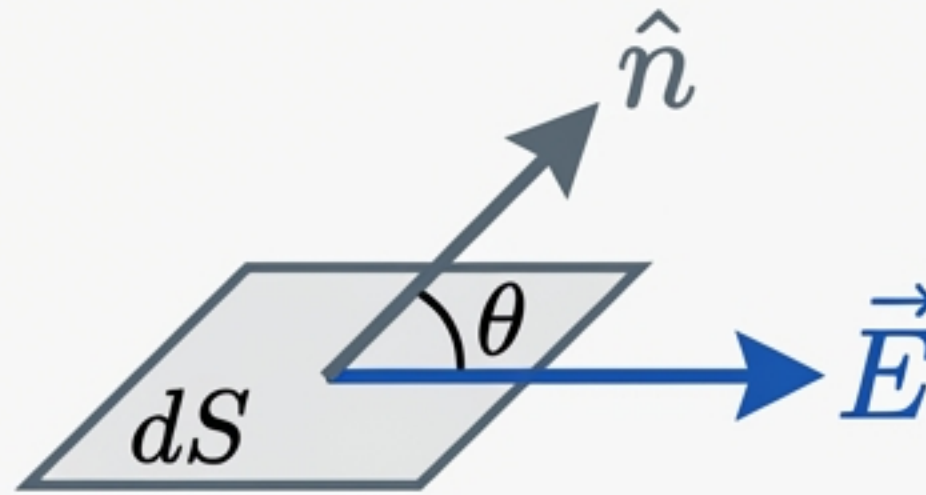
Definition & Convention

$$d\vec{S} = dS \hat{n}$$

Convention: For closed surfaces, the area vector always points along the **outward normal**.

Calculating Electric Flux

$$\Delta\Phi = \vec{E} \cdot d\vec{S} = E dS \cos \theta$$

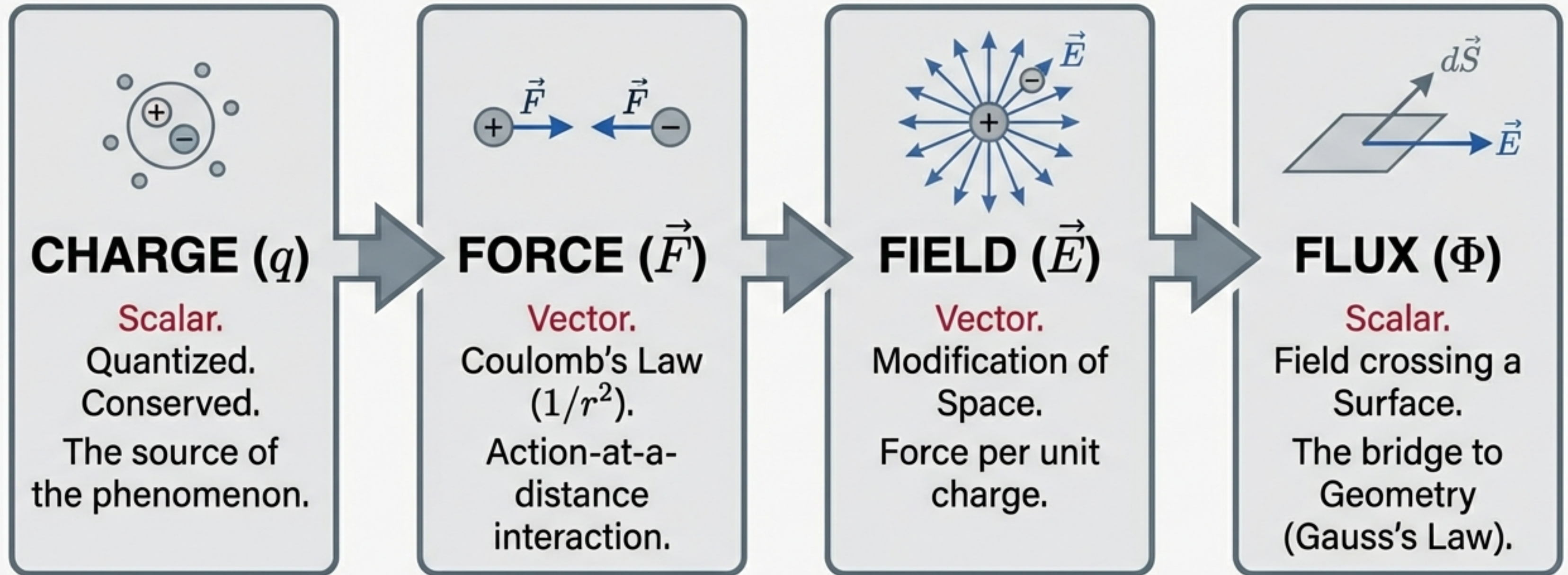


$\theta = 0^\circ$ (Normal $\parallel$ Field):
Flux = $+E dS$ (Max).

$\theta = 90^\circ$ (Surface $\parallel$ Field):
Flux = 0.

Total Flux (Integration):
 $\Phi \approx \sum \vec{E} \cdot d\vec{S}.$

Summary: The Electrostatic Progression



This logical chain—from particle to spatial model—establishes the foundation for all Electromagnetic Theory.